Element H

High Priority Design Requirements

1. Safety:
    a. Must incite 0 total safety hazards for the crew whilst in operation, either by the method of power, operation, or form.

Test: We will test the viability of our design with regard to the number of injuries caused by the mechanism by measuring the numbering of injuries with a goal of zero as the minimum for success. The test will be conducted by running through every use of the prop to test for any malfunctions. Setting up and using the prop including moving the curtain, changing or adding batteries, wounding up the string and releasing it. First start by flipping the brown curtain up in order to access the batteries and motors. The batteries are in the black box and can be accessed by removing the top with pressure. Then take the batteries out and replace them with new ones. Next move onto the wound up the string by holding the petal where you want it and use the controller by moving the joystick labeled with that petal. Once in palace, push the joystick in the opposite direction. Repeat this seven times and record all injuries caused by this mechanism. The expected range of injuries from using this device is zero. The minimum result that would meet the design requirement is zero. Our team's goal is to have zero injuries caused by using this device. The dependent variable is the number of injuries and the independent variable is how the device is being used. The controls will be controlled by the stage crew person responsible for the rose, amount of light, the time and steps of setup.

2. Petals Fall:
    a. Must have at least 3 instances where one or more petals fall at specific times set/initiated by the tech crew of the musical.

Test: We will test the viability of our design with regard to the number of petals to fall by measuring numbers of petals that fall with a goal of three petals as the minimum for success. The test will be set up by wounding up the string and then releasing it via moving the joysticks on the controller in the labeled direction. During the test, all three petals need to drop. The expected range of results that could be recorded each test is zero to three petals. The minimum result that is required is three petals.This step should be repeated seven times and recorded. The controls are the petals will start wounding up, the joystick pressed the same each time, the order of petals released each time and the time in between dropping each petal. The dependent variable is the number of petals that fall and the independent variable is how the petals are released.

3. Reliability:
    a. Must sustain 0 mechanical or operational issues or damages for at least 10 uses.

Test: We will test the viability of our design with regard to the number of operational or mechanical issues or damages by measuring the number of operational issues or damages with a goal of zero as the minimum for success. To test this, the wounding up of the string and dropping the petals should be done 10 times and each error recorded. The expected range of results is zero to ten errors. The minimum result to meet the design required is zero errors. The controls are set up procedure, stage crew member in charge of prop and lighting. The dependent variable is the number of operational or mechanical issues or damages and the independent variable is the number of times it is used.

4. Power:
    a. Must have access to at least 60v of power.

Test: We will test the viability of our design with regard to the amount of power by measuring the amount of voltages of power with a goal of 60v as the minimum for success. To test this, we will use a digital multimeter to measure the voltages during use. We will run the prop as intended and record the voltages as it is used. Then we will reset it and repeat the test seven times. The minimum result to meet the design requirement is 60 volts. The controls will be set up, placement of digital multimeter, duration of use, type of batteries and usage of batteries. The dependent variable is the voltage and the independent variable is materials that were used to create the mechanism.

5. Structure:
    a. Must stay on the pedestal, and not lean more than 0 degrees for the entirety of the play.

Test: We will test the viability of our design with regard to structure by measuring the degrees in tilt with a goal of zero as the maximum for success. To set up the test, we will set up the rose to be used then measure the angle. Then we'll use the mechanism following the play's time length and release time. After, we'll record the degrees of tilt then repeat seven times. During the play there needs to be zero degrees in tilt. The minimum result to meet the design requirement is zero degrees in tilt. The controls of the test are set up, duration, release timing, and platform the mechanism sits on. The dependent variable is the degrees in tilt and the independent variable is how long it is used for.

Low Priority Design Requirements

6. Lights:
    a. Must have at least 3 individual lights aimed at the rose from below the bulb.
7. Color:
    a. Must have red petals and a green stem.
8. Hight:
    a. Must be at least 1.5ft tall from the base of the flower to the top of the flower.
9. Cloche
    a. Must not be enclosed by a cloche with an internal volume less than the volume of the rose
10. Weight:
    a. Must be lighter than 10lbs.